Vendor_Performance_Analysis

February 27, 2026

[95]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
import sqlite3
from scipy.stats import ttest_ind
import scipy.stats as stats
warnings.filterwarnings('ignore')
```

1 Load Data Set

[96]:
```
# Creating the connection
conn = sqlite3.connect('inventory.db')


# Fetching vendor summary data
df = pd.read_sql_query("""  SELECT * FROM vendor_sales_summary""", conn)
df.head()
```

[96]:
```
   VendorNumber                VendorName  Brand              Description  \
0          1128         BROWN-FORMAN CORP   1233  Jack Daniels No 7 Black
1          4425     MARTIGNETTI COMPANIES   3405    Tito's Handmade Vodka
2         17035         PERNOD RICARD USA   8068         Absolut 80 Proof
3          3960  DIAGEO NORTH AMERICA INC   4261   Capt Morgan Spiced Rum
4          3960  DIAGEO NORTH AMERICA INC   3545          Ketel One Vodka

   PurchasePrice  ActualPrice  Volume  TotalPurchaseQuantity  \
0          26.27        36.99  1750.0                 145080
1          23.19        28.99  1750.0                 164038
2          18.24        24.99  1750.0                 187407
3          16.17        22.99  1750.0                 201682
4          21.89        29.99  1750.0                 138109

   TotalPurchaseDollars  TotalSalesQuantity  TotalSalesDollars  \
0            3811251.60            142049.0         5101919.51
1            3804041.22            160247.0         4819073.49
```

```
2            3418303.68            187140.0         4538120.60
3            3261197.94            200412.0         4475972.88
4            3023206.01            135838.0         4223107.62

   TotalSalesPrice  ExciseTax  FreightCost  GrossProfit  ProfitMargin  \
0        672819.31  260999.20     68601.68   1290667.91     25.297693
1        561512.37  294438.66    144929.24   1015032.27     21.062810
2        461140.15  343854.07    123780.22   1119816.92     24.675786
3        420050.01  368242.80    257032.07   1214774.94     27.139908
4        545778.28  249587.83    257032.07   1199901.61     28.412764

   SalesTurnover  SalestoPurchaseRatio
0       0.979108              1.338647
1       0.976890              1.266830
2       0.998575              1.327594
3       0.993703              1.372493
4       0.983556              1.396897
```

2 Exploratory Data Analysis

- In the previous step we reviewed the database schema and table relationships to identify candidate variables and the tables to join for the final dataset.

- In this EDA phase we will profile the resulting joined table column-by-column to quantify distributions, detect outliers and invalid values, and measure data quality (missingness, duplicates, type mismatches, and inconsistent categories).

- Outputs will include: summary statistics by column (count, mean/median, IQR, min/max), frequency tables for categorical fields, datetime range checks, missing-value and duplicate reports, and a prioritized list of anomalies requiring remediation before modelling or reporting.

- Acceptance criteria for moving forward: no critical data-type mismatches, all high-impact missing values either explained or imputed with documented rationale, and key outliers validated or flagged for exclusion.

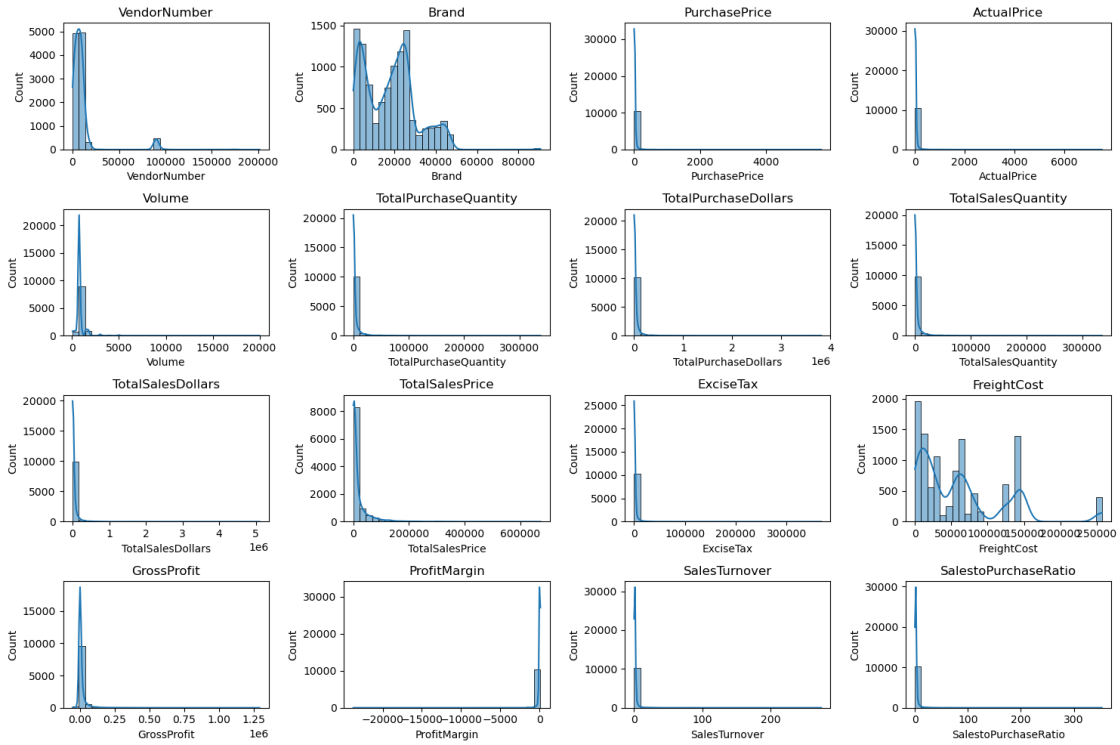
[97]:
```python
# Summary Statistics
df.describe().T
```

[97]:
```
                         count          mean            std           min  \
VendorNumber           10692.0  10650.649458   18753.519148      2.000000
Brand                  10692.0  18039.228769   12662.187074     58.000000
PurchasePrice          10692.0     24.385303     109.269375      0.360000
ActualPrice            10692.0     35.643671     148.246016      0.490000
Volume                 10692.0    847.360550     664.309212     50.000000
TotalPurchaseQuantity  10692.0   3140.886831   11095.086769      1.000000
TotalPurchaseDollars   10692.0  30106.693372  123067.799627      0.710000
TotalSalesQuantity     10514.0   3129.583317   11037.801407      1.000000
```

```
TotalSalesDollars      10514.0  42954.173834  168977.755843      1.980000
TotalSalesPrice        10514.0  19111.958773   45264.605387      0.990000
ExciseTax              10514.0   1804.263569   11065.659557      0.060000
FreightCost            10692.0  61433.763214   60938.458032      0.090000
GrossProfit            10514.0  12364.618776   46576.616345 -52002.780000
ProfitMargin           10514.0    -15.885227     447.289882 -23730.638953
SalesTurnover          10514.0      1.735689       6.067081      0.002817
SalestoPurchaseRatio   10514.0      2.546789       8.524047      0.004196

                                25%           50%           75%           max
VendorNumber            3951.000000   7153.000000   9552.000000  2.013590e+05
Brand                   5793.500000  18761.500000  25514.250000  9.063100e+04
PurchasePrice              6.840000     10.455000     19.482500  5.681810e+03
ActualPrice               10.990000     15.990000     28.990000  7.499990e+03
Volume                   750.000000    750.000000    750.000000  2.000000e+04
TotalPurchaseQuantity     36.000000    262.000000   1975.750000  3.376600e+05
TotalPurchaseDollars     453.457500   3655.465000  20738.245000  3.811252e+06
TotalSalesQuantity        36.000000    280.000000   1985.000000  3.349390e+05
TotalSalesDollars        809.820000   5599.700000  29524.250000  5.101920e+06
TotalSalesPrice          334.890000   3020.085000  16442.477500  6.728193e+05
ExciseTax                  5.382500     50.425000    436.730000  3.682428e+05
FreightCost            14069.870000  50293.620000  79528.990000  2.570321e+05
GrossProfit               66.975000   1496.490000   8970.040000  1.290668e+06
ProfitMargin              15.353839     30.778375     40.210967  9.971666e+01
SalesTurnover              0.829761      0.983431      1.042183  2.745000e+02
SalestoPurchaseRatio       1.181388      1.444635      1.672548  3.529286e+02
```

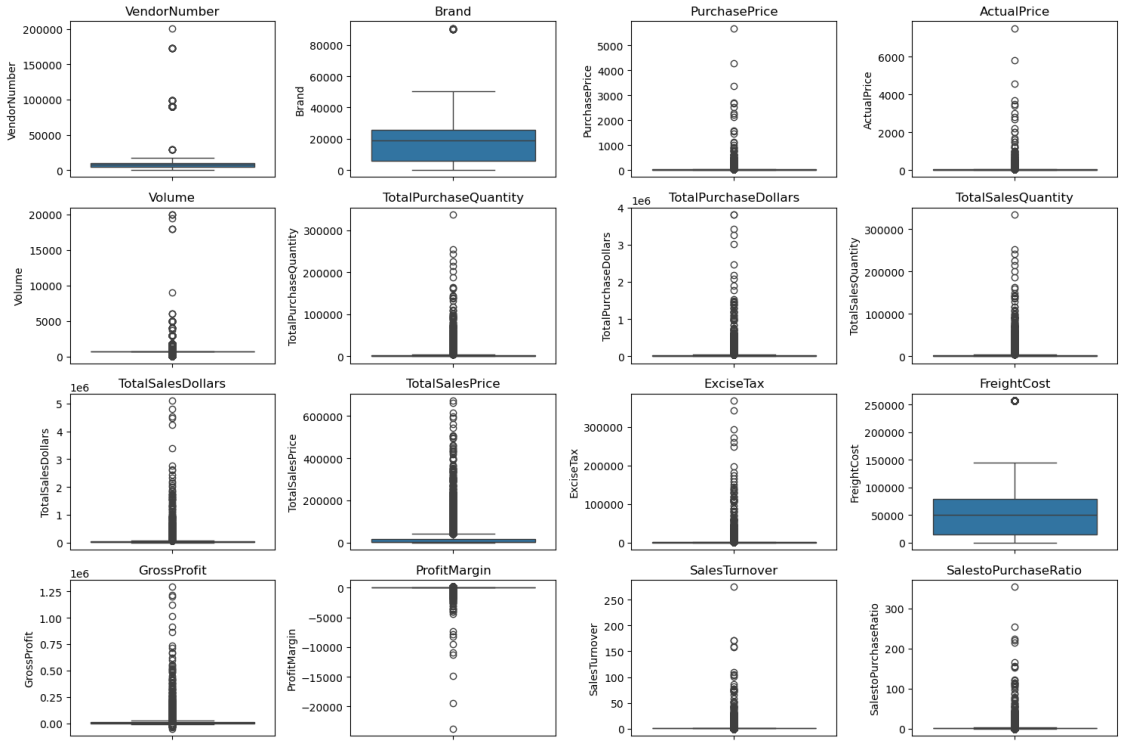
[98]:
```python
# Distribution Columns for Numerical Columns

numerical_cols = df.select_dtypes(include=np.number).columns

plt.figure(figsize=(15,10))
for i, col in enumerate(numerical_cols):
    plt.subplot(4,4, i+1)   # Adjust layout as needed
    sns.histplot(df[col],kde=True , bins=30)
    plt.title(col)
plt.tight_layout()
plt.show
```

[98]:
```
<function matplotlib.pyplot.show(close=None, block=None)>
```



[99]:
```
# Outlier detection with box plots

plt.figure(figsize=(15,10))
for i, col in enumerate(numerical_cols):
    plt.subplot(4,4, i+1)
    sns.boxplot(y=df[col])
    plt.title(col)
plt.tight_layout()
plt.show
```

[99]: <function matplotlib.pyplot.show(close=None, block=None)>



2.0.1 Summary Statistics Insights

Negative & Zero Values

- **Gross Profit = -5,202.78**, indicating losses. Some products or transactions may be selling at a loss due to high costs or selling at discounts lower than the purchase price.

- **Profit Margin: minimum = -∞**, which suggests cases where revenue is zero or even lower than costs.

- **Total Sales Quantity & Sales Dollars: minimum = 0**, meaning some products were purchased but never sold. These could be slow-moving or obsolete stock.

Outliers Indicated by High Standard Deviations

- **Purchase & Actual Prices**: max values (5,681.81 and 7,499.99) are significantly higher than the mean (24.39 and 3.56A), indicating potential premium products or data entry errors.

- **Freight Cost**: huge variation from 0.09 to 257,032.07, suggesting logistics inefficiencies or bulk shipments.

- **Stock Turn**: ranges from 0 to 274.5, implying some products sell extremely fast while others remain in stock indefinitely. Values greater than 1 indicate the product is sold more than

purchased quantity, possibly because sales are being fulfilled from other stock.

[100]:
```
# Filtering the data by removing inconsistencies
df = pd.read_sql_query("""SELECT *
FROM vendor_sales_summary
WHERE GrossProfit > 0
AND ProfitMargin > 0
AND TotalSalesQuantity > 0 """,conn)
```

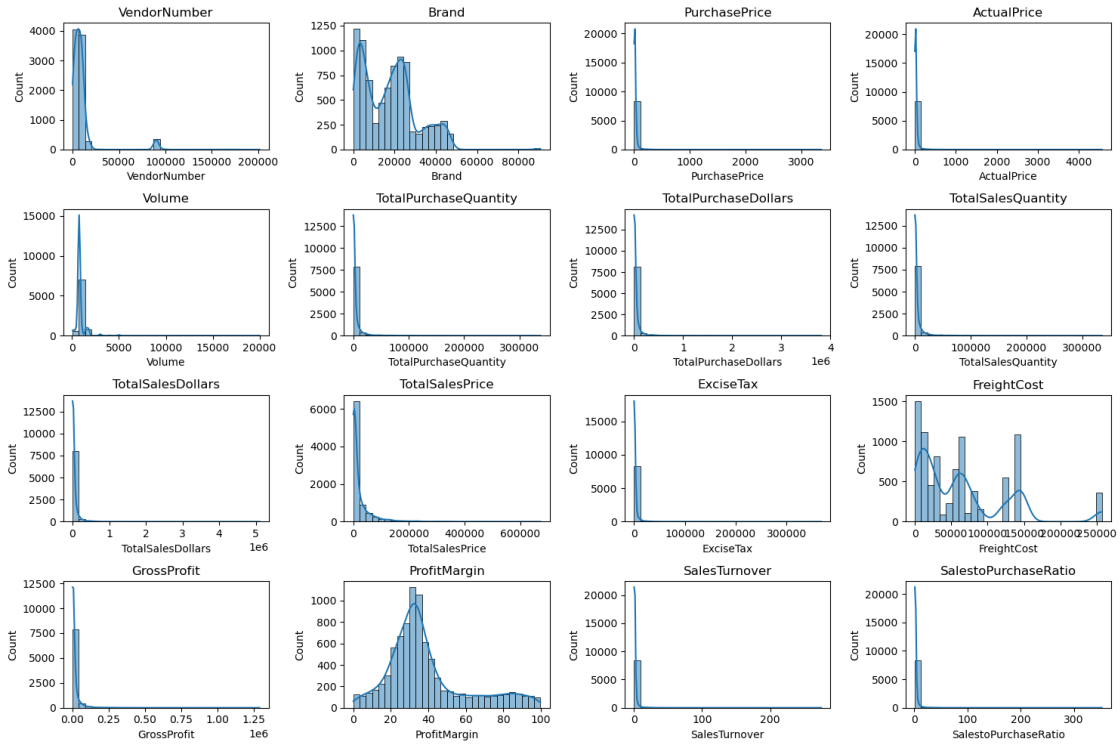
[101]:
```
df
```

[101]:
```
      VendorNumber                VendorName  Brand  \
0             1128         BROWN-FORMAN CORP   1233
1             4425     MARTIGNETTI COMPANIES   3405
2            17035         PERNOD RICARD USA   8068
3             3960  DIAGEO NORTH AMERICA INC   4261
4             3960  DIAGEO NORTH AMERICA INC   3545
…             …                        …    …
8559          9815            WINE GROUP INC   8527
8560          8004            SAZERAC CO INC   5683
8561          3924  HEAVEN HILL DISTILLERIES   9123
8562          3960  DIAGEO NORTH AMERICA INC   6127
8563          7245      PROXIMO SPIRITS INC.   3065

                      Description  PurchasePrice  ActualPrice  Volume  \
0         Jack Daniels No 7 Black          26.27        36.99  1750.0
1           Tito's Handmade Vodka          23.19        28.99  1750.0
2                Absolut 80 Proof          18.24        24.99  1750.0
3          Capt Morgan Spiced Rum          16.17        22.99  1750.0
4                 Ketel One Vodka          21.89        29.99  1750.0
…                             …            …           …      …
8559  Concannon Glen Ellen Wh Zin           1.32         4.99   750.0
8560  Dr McGillicuddy's Apple Pie           0.39         0.49    50.0
8561              Deep Eddy Vodka           0.74         0.99    50.0
8562  The Club Strawbry Margarita           1.47         1.99   200.0
8563     Three Olives Grape Vodka           0.71         0.99    50.0

      TotalPurchaseQuantity  TotalPurchaseDollars  TotalSalesQuantity  \
0                    145080            3811251.60            142049.0
1                    164038            3804041.22            160247.0
2                    187407            3418303.68            187140.0
3                    201682            3261197.94            200412.0
4                    138109            3023206.01            135838.0
…                     …                    …                   …
8559                      2                  2.64                 5.0
8560                      6                  2.34               134.0
8561                      2                  1.48                 2.0
```

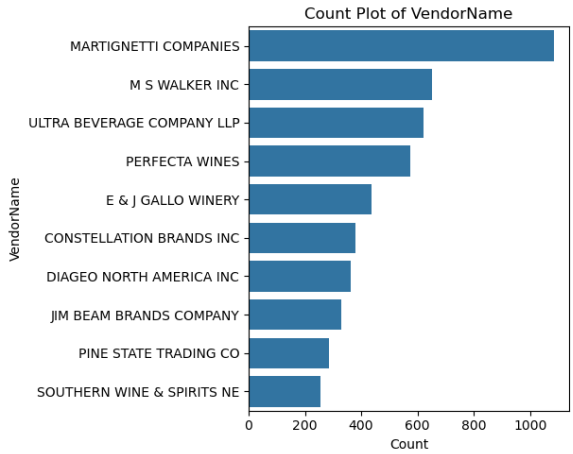
```
8562                      1                  1.47                72.0
8563                      1                  0.71                86.0

      TotalSalesDollars  TotalSalesPrice  ExciseTax  FreightCost  GrossProfit  \
0            5101919.51        672819.31  260999.20     68601.68   1290667.91
1            4819073.49        561512.37  294438.66    144929.24   1015032.27
2            4538120.60        461140.15  343854.07    123780.22   1119816.92
3            4475972.88        420050.01  368242.80    257032.07   1214774.94
4            4223107.62        545778.28  249587.83    257032.07   1199901.61
…                 …                 …          …           …           …
8559              15.95             10.96       0.55     27100.41        13.31
8560              65.66              1.47       7.04     50293.62        63.32
8561               1.98              0.99       0.10     14069.87         0.50
8562             143.28             77.61      15.12    257032.07       141.81
8563              85.14             33.66       4.46     38994.78        84.43

      ProfitMargin  SalesTurnover  SalestoPurchaseRatio
0        25.297693       0.979108              1.338647
1        21.062810       0.976890              1.266830
2        24.675786       0.998575              1.327594
3        27.139908       0.993703              1.372493
4        28.412764       0.983556              1.396897
…            …             …                    …
8559     83.448276       2.500000              6.041667
8560     96.436186      22.333333             28.059829
8561     25.252525       1.000000              1.337838
8562     98.974037      72.000000             97.469388
8563     99.166079      86.000000            119.915493

[8564 rows x 18 columns]
```

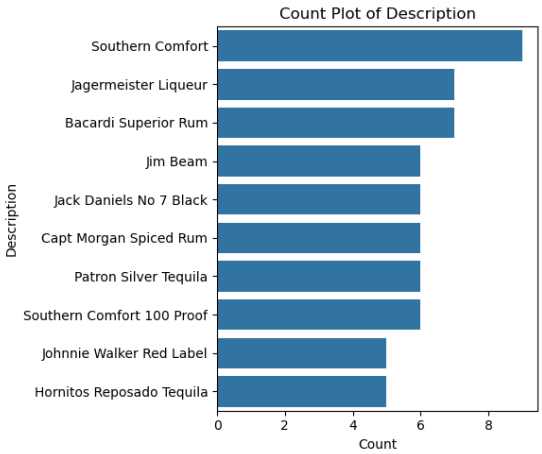
[102]:
```
# Distribution Columns for Numerical Columns after cleaning

numerical_cols = df.select_dtypes(include=np.number).columns

plt.figure(figsize=(15,10))
for i, col in enumerate(numerical_cols):
    plt.subplot(4,4, i+1)   # Adjust layout as needed
    sns.histplot(df[col],kde=True , bins=30)
    plt.title(col)
plt.tight_layout()
plt.show
```

[102]:
```
<function matplotlib.pyplot.show(close=None, block=None)>
```

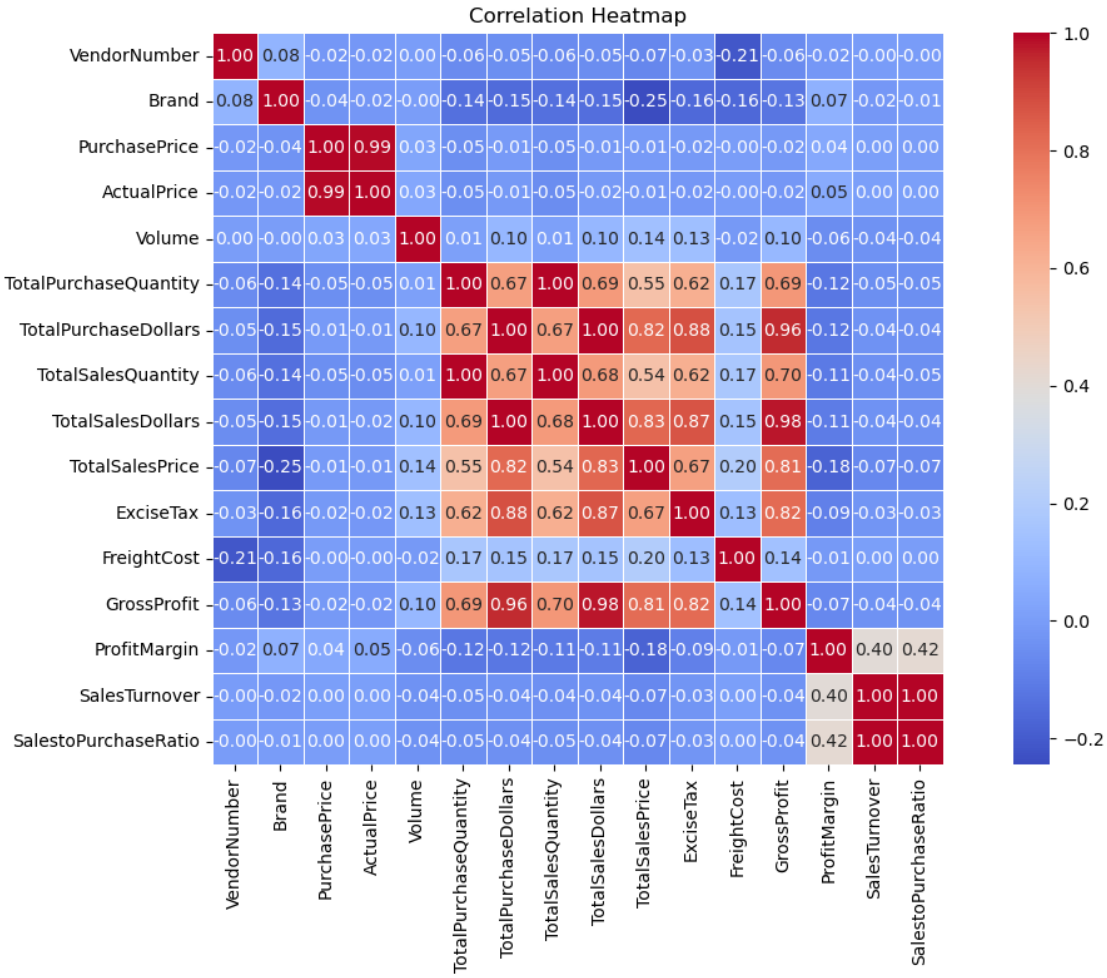


[103]:
```
# count plot for categrical columns
import matplotlib.pyplot as plt
import seaborn as sns

categorical_cols = ['VendorName', 'Description']

plt.figure(figsize=(12, 5))
for i, col in enumerate(categorical_cols):
    plt.subplot(1, 2, i + 1)
    top10 = df[col].value_counts().index[:10]
    sns.countplot(y=col, data=df, order=top10)
    plt.title(f"Count Plot of {col}")
    plt.xlabel("Count")
    plt.ylabel(col)
plt.tight_layout()
plt.show()
```





[104]:
```
# correlation heatmap
plt.figure(figsize=(12, 8))
correlation_matrix = df[numerical_cols].corr()
sns.heatmap(correlation_matrix, annot=True, fmt=".2f", cmap="coolwarm",␣
↪linewidths=0.5, square=True)
plt.title("Correlation Heatmap")
plt.tight_layout()
plt.show()
```



2.0.2 Correlation Insights

- PurchasePrice has weak correlations with TotalSalesDollars (-0.012) and GrossProfit (-0.016), suggesting that price variations do not significantly impact sales revenue or profit.

- Strong correlation between total purchase quantity and total sales quantity (0.999), confirming efficient inventory turnover.

- Negative correlation between profit margin & total sales price (-0.179) suggests that as sales price increases, margins decrease, possibly due to competitive pricing pressures.

- StockTurnover has weak negative correlations with both GrossProfit (-0.038) and ProfitMargin (-0.055), indicating that faster turnover does not necessarily result in higher profitability.

2.1 Data Analysis

Identify the Brands that need Pricing or Pricing adjustment which exibits lower sales performace but higher profit margin

[105]:
```
brand_performance = df.groupby('Description').agg({
    'TotalSalesDollars':'sum',
    'ProfitMargin':'mean'}).reset_index()
```

[106]:
```
low_sales_threshold = brand_performance['TotalSalesDollars'].quantile(0.15)
high_margin_threshold = brand_performance['ProfitMargin'].quantile(0.85)
```

[107]:
```
low_sales_threshold
```

[107]:
```
np.float64(560.299)
```

[108]:
```
high_margin_threshold
```

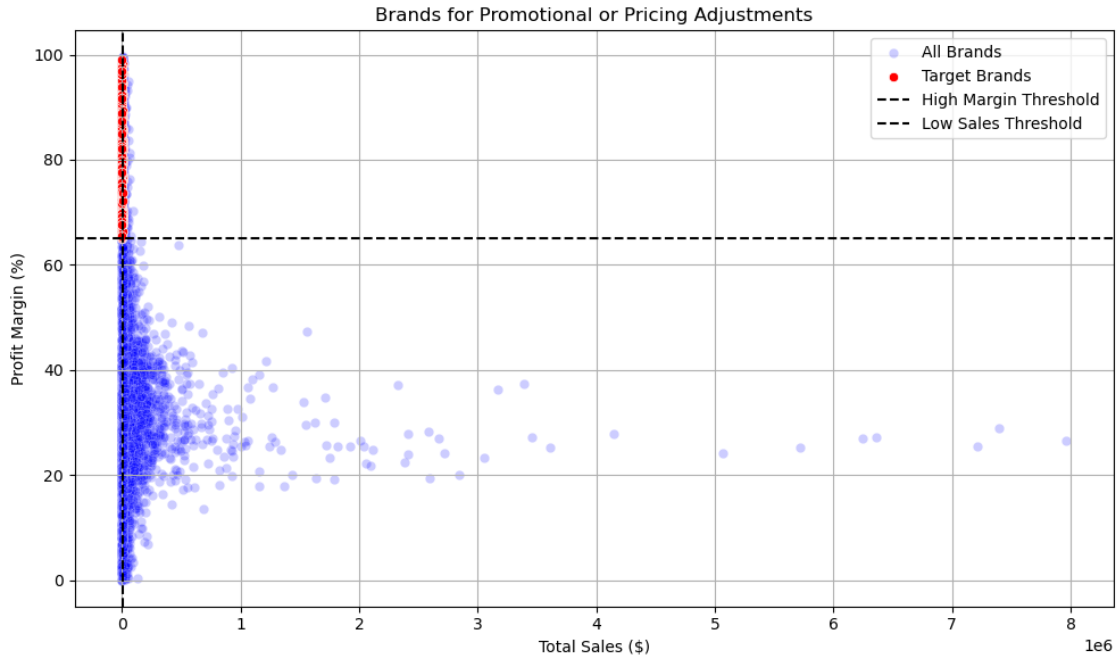
[108]:
```
np.float64(64.97017552750113)
```

[109]:
```
# Filtering brand with low sales but high profit margin
target_brands = brand_performance[
    (brand_performance['TotalSalesDollars'] <= low_sales_threshold)&
    (brand_performance['ProfitMargin'] >= high_margin_threshold)
]

print("Brands with low sales but high profit margin:")
display(target_brands.sort_values('TotalSalesDollars'))
```

```
Brands with low sales but high profit margin:

                       Description  TotalSalesDollars  ProfitMargin
6199    Santa Rita Organic Svgn Bl               9.99     66.466466
2369             Debauchery Pnt Nr              11.58     65.975820
2070   Concannon Glen Ellen Wh Zin              15.95     83.448276
2188             Crown Royal Apple              27.86     89.806174
6237  Sauza Sprklg Wild Berry Marg              27.96     82.153076
…                              …                …            …
5074   Nanbu Bijin Southern Beauty             535.68     76.747312
2271         Dad's Hat Rye Whiskey             538.89     81.851584
57       A Bichot Clos Marechaudes             539.94     67.740860
6245     Sbragia Home Ranch Merlot             549.75     66.444748
3326     Goulee Cos d'Estournel 10             558.87     69.434752

[198 rows x 3 columns]
```

[110]:
```
plt.figure(figsize=(10, 6))
sns.scatterplot(
    data=brand_performance,
```

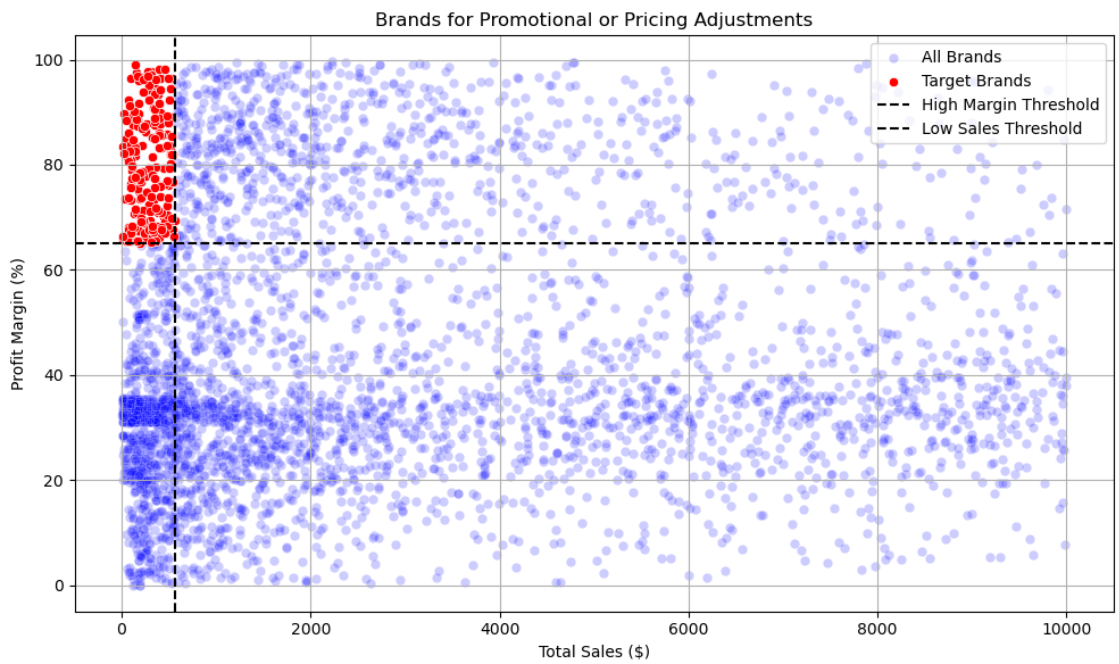
```
    x='TotalSalesDollars',
    y='ProfitMargin',
    color="blue",
    label="All Brands",
    alpha=0.2
)
sns.scatterplot(
    data=target_brands,
    x='TotalSalesDollars',
    y='ProfitMargin',
    color="red",
    label="Target Brands",
    edgecolor='w',
    linewidth=0.5
)

# Reference lines: horizontal for margin, vertical for sales
plt.axhline(y=high_margin_threshold, linestyle='--', color='black', label="High␣
 ↪Margin Threshold")
plt.axvline(x=low_sales_threshold, linestyle='--', color='black', label="Low␣
 ↪Sales Threshold")

plt.xlabel("Total Sales ($)")
plt.ylabel("Profit Margin (%)")
plt.title("Brands for Promotional or Pricing Adjustments")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



[111]:
```
brand_performance = brand_performance[brand_performance['TotalSalesDollars'] <␣
↪10000]  # for better visulisation
```

[112]:
```
plt.figure(figsize=(10, 6))
sns.scatterplot(
    data=brand_performance,
    x='TotalSalesDollars',
    y='ProfitMargin',
    color="blue",
    label="All Brands",
    alpha=0.2
)
sns.scatterplot(
    data=target_brands,
    x='TotalSalesDollars',
    y='ProfitMargin',
    color="red",
    label="Target Brands",
    edgecolor='w',
    linewidth=0.5
)

# Reference lines: horizontal for margin, vertical for sales
plt.axhline(y=high_margin_threshold, linestyle='--', color='black', label="High␣
↪Margin Threshold")
```

```
plt.axvline(x=low_sales_threshold, linestyle='--', color='black', label="Low␣
↪Sales Threshold")

plt.xlabel("Total Sales ($)")
plt.ylabel("Profit Margin (%)")
plt.title("Brands for Promotional or Pricing Adjustments")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



Which Vendors and Brands represent the higest sales performace ?

[113]:
```
def format_dollars(value):
    if value >= 1_000_000:
        return f"{value / 1_000_000:.2f}M"
    elif value >= 1_000:
        return f"{value / 1_000:.2f}K"
    else:
        return str(value)
```

[114]:
```
# Top Vendors & Brands by Sales Performance
top_vendors = df.groupby("VendorName")["TotalSalesDollars"].sum().nlargest(10)
top_brands = df.groupby("Description")["TotalSalesDollars"].sum().nlargest(10)

print("Top 10 Vendors by Total Sales:")
```

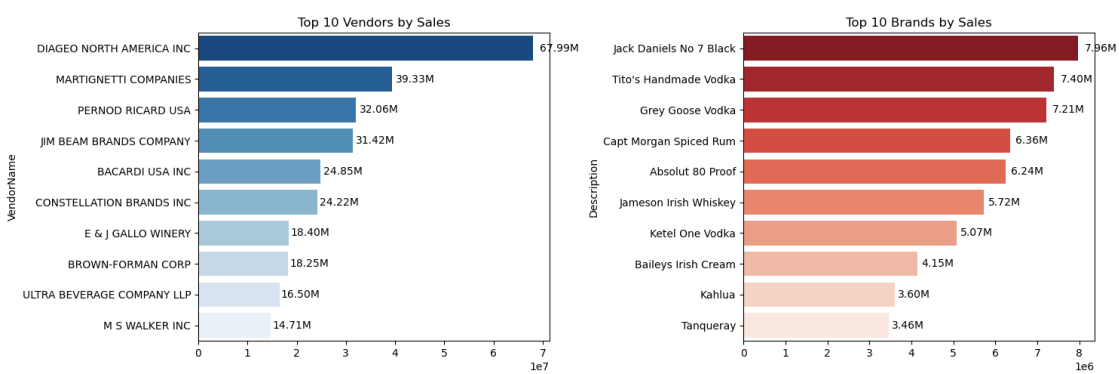
```python
print(top_vendors)

print("\nTop 10 Brands by Total Sales:")
print(top_brands)
```

```
Top 10 Vendors by Total Sales:
VendorName
DIAGEO NORTH AMERICA INC      67990099.42
MARTIGNETTI COMPANIES         39330359.36
PERNOD RICARD USA             32063196.19
JIM BEAM BRANDS COMPANY       31423020.46
BACARDI USA INC               24854817.14
CONSTELLATION BRANDS INC      24218745.65
E & J GALLO WINERY            18399899.46
BROWN-FORMAN CORP             18247230.65
ULTRA BEVERAGE COMPANY LLP    16502544.31
M S WALKER INC                14706458.51
Name: TotalSalesDollars, dtype: float64

Top 10 Brands by Total Sales:
Description
Jack Daniels No 7 Black    7964746.76
Tito's Handmade Vodka      7399657.58
Grey Goose Vodka           7209608.06
Capt Morgan Spiced Rum     6356320.62
Absolut 80 Proof           6244752.03
Jameson Irish Whiskey      5715759.69
Ketel One Vodka            5070083.56
Baileys Irish Cream        4150122.07
Kahlua                     3604858.66
Tanqueray                  3456697.90
Name: TotalSalesDollars, dtype: float64
```

[115]:
```python
top_brands.apply(lambda x: format_dollars(x))
```

[115]:
```
Description
Jack Daniels No 7 Black    7.96M
Tito's Handmade Vodka      7.40M
Grey Goose Vodka           7.21M
Capt Morgan Spiced Rum     6.36M
Absolut 80 Proof           6.24M
Jameson Irish Whiskey      5.72M
Ketel One Vodka            5.07M
Baileys Irish Cream        4.15M
Kahlua                     3.60M
Tanqueray                  3.46M
Name: TotalSalesDollars, dtype: object
```

[116]:
```
plt.figure(figsize=(15,5))

# PLot for top vendors
plt.subplot(1,2,1)
ax1 = sns.barplot( x= top_vendors.values, y= top_vendors.index, palette␣
↪="Blues_r")
plt.title("Top 10 Vendors by Sales")

for bar in ax1.patches:
    ax1.text(bar.get_width() + (bar.get_width() * 0.02),
             bar.get_y() + bar.get_height() / 2,
             format_dollars(bar.get_width()),
             ha='left', va='center', fontsize=10, color='black')

# Plot for Top Brands
ax2 = plt.subplot(1, 2, 2)
ax2 = sns.barplot(y=top_brands.index.astype(str), x=top_brands.values,␣
↪palette="Reds_r")
ax2.set_title("Top 10 Brands by Sales")

for bar in ax2.patches:
    ax2.text(bar.get_width() + (bar.get_width() * 0.02),
             bar.get_y() + bar.get_height() / 2,
             format_dollars(bar.get_width()),
             ha='left', va='center', fontsize=10, color='black')

plt.tight_layout()
plt.show()
```



Which Vendor contributes the most to total purcahse dollars ?

[117]:
```
vendor_performance =df.groupby('VendorName').agg({
    'TotalPurchaseDollars':'sum',
    'GrossProfit':'sum',
```

```
    'TotalSalesDollars':'sum'
}).reset_index()
vendor_performance.shape
```

[117]: (119, 4)

[118]:
```
vendor_performance['Purchase_Contribution%'] =␣
↪vendor_performance['TotalPurchaseDollars']/
↪vendor_performance['TotalPurchaseDollars'].sum()*100
```

[119]:
```
vendor_performance
```

[119]:
```
                      VendorName  TotalPurchaseDollars  GrossProfit  \
0        ADAMBA IMPORTS INTL INC                446.16       258.37
1           ALISA CARR BEVERAGES              25698.12     78772.82
2             ALTAMAR BRANDS LLC              11706.20      4000.61
3      AMERICAN SPIRITS EXCHANGE                934.08       577.08
4      AMERICAN VINTAGE BEVERAGE             104435.68     35167.85
..                           …                     …           …
114               WEIN BAUER INC              42694.64     13522.49
115  WESTERN SPIRITS BEVERAGE CO             298416.86    106837.97
116     WILLIAM GRANT & SONS INC            5876538.26   1693337.94
117               WINE GROUP INC            5203801.17   3100242.11
118            ZORVINO VINEYARDS              86122.71     38066.88

     TotalSalesDollars  Purchase_Contribution%
0               704.53                0.000145
1            104470.94                0.008362
2             15706.81                0.003809
3              1511.16                0.000304
4            139603.53                0.033981
..                 …                       …
114           56217.13                0.013892
115          405254.83                0.097097
116         7569876.20                1.912074
117         8304043.28                1.693183
118          124189.59                0.028022

[119 rows x 5 columns]
```

[120]:
```
vendor_performance = round(vendor_performance.
↪sort_values('Purchase_Contribution%',ascending=False),2)
```

[121]:
```
# Display Top 10 Vendors
top_vendors = vendor_performance.head(10)
top_vendors['TotalSalesDollars'] = top_vendors['TotalSalesDollars'].
↪apply(format_dollars)
```

```
top_vendors['TotalPurchaseDollars'] = top_vendors['TotalPurchaseDollars'].
↪apply(format_dollars)
top_vendors['GrossProfit'] = top_vendors['GrossProfit'].apply(format_dollars)
top_vendors
```

[121]:

```
                     VendorName TotalPurchaseDollars GrossProfit  \
25     DIAGEO NORTH AMERICA INC               50.10M      17.89M
57        MARTIGNETTI COMPANIES               25.50M      13.83M
68            PERNOD RICARD USA               23.85M       8.21M
46      JIM BEAM BRANDS COMPANY               23.49M       7.93M
6               BACARDI USA INC               17.43M       7.42M
20     CONSTELLATION BRANDS INC               15.27M       8.95M
11            BROWN-FORMAN CORP               13.24M       5.01M
30           E & J GALLO WINERY               12.07M       6.33M
106  ULTRA BEVERAGE COMPANY LLP               11.17M       5.34M
53               M S WALKER INC                9.76M       4.94M

    TotalSalesDollars  Purchase_Contribution%
25             67.99M                   16.30
57             39.33M                    8.30
68             32.06M                    7.76
46             31.42M                    7.64
6              24.85M                    5.67
20             24.22M                    4.97
11             18.25M                    4.31
30             18.40M                    3.93
106            16.50M                    3.63
53             14.71M                    3.18
```

[122]:

```
top_vendors['Purchase_Contribution%'].sum()
```

[122]: np.float64(65.69000000000001)

[123]:

```
top_vendors['Cumulative_Contribution%'] = top_vendors['Purchase_Contribution%'].
↪cumsum()
top_vendors
```

[123]:

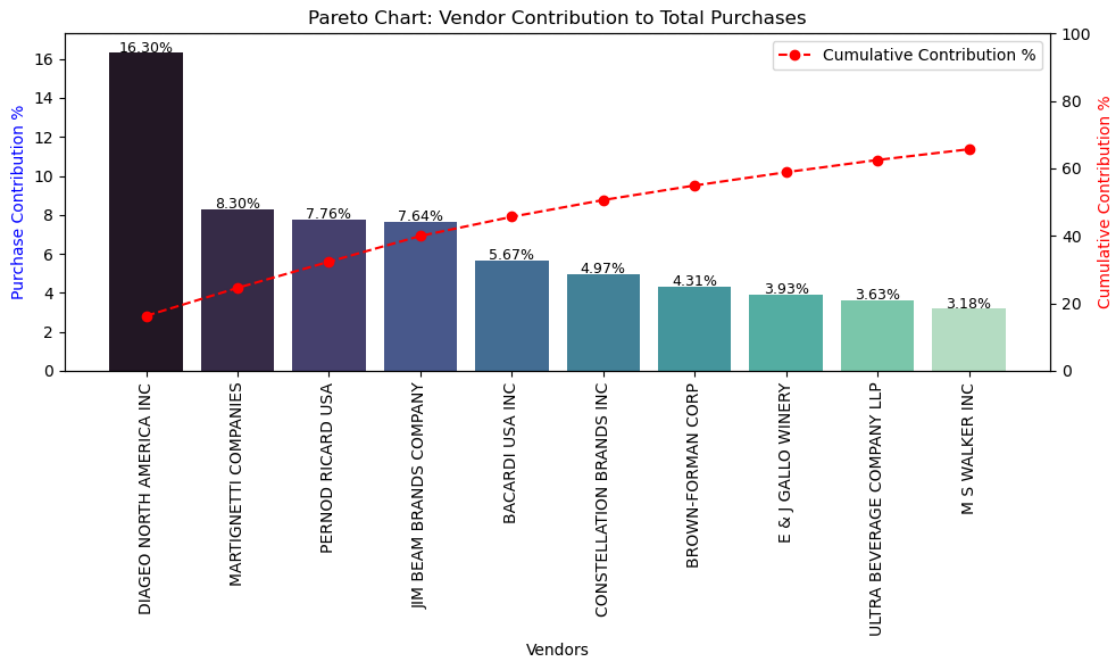
```
                     VendorName TotalPurchaseDollars GrossProfit  \
25     DIAGEO NORTH AMERICA INC               50.10M      17.89M
57        MARTIGNETTI COMPANIES               25.50M      13.83M
68            PERNOD RICARD USA               23.85M       8.21M
46      JIM BEAM BRANDS COMPANY               23.49M       7.93M
6               BACARDI USA INC               17.43M       7.42M
20     CONSTELLATION BRANDS INC               15.27M       8.95M
11            BROWN-FORMAN CORP               13.24M       5.01M
30           E & J GALLO WINERY               12.07M       6.33M
106  ULTRA BEVERAGE COMPANY LLP               11.17M       5.34M
```

```
53                M S WALKER INC                 9.76M       4.94M

    TotalSalesDollars  Purchase_Contribution%  Cumulative_Contribution%
25             67.99M                   16.30                     16.30
57             39.33M                    8.30                     24.60
68             32.06M                    7.76                     32.36
46             31.42M                    7.64                     40.00
6              24.85M                    5.67                     45.67
20             24.22M                    4.97                     50.64
11             18.25M                    4.31                     54.95
30             18.40M                    3.93                     58.88
106            16.50M                    3.63                     62.51
53             14.71M                    3.18                     65.69
```

[138]:

```python
import matplotlib.pyplot as plt
import seaborn as sns

fig, ax1 = plt.subplots(figsize=(10, 6))

# Bar plot for Purchase Contribution %
sns.barplot(x=top_vendors['VendorName'],
            y=top_vendors['Purchase_Contribution%'],
            palette="mako", ax=ax1)

# Add bar labels slightly above each bar
for i, value in enumerate(top_vendors['Purchase_Contribution%']):
    ax1.text(i, value + 0.05, f"{value:.2f}%", ha='center', fontsize=9,␣
 ↪color='black')

# Line Plot for Cumulative Contribution %
ax2 = ax1.twinx()
ax2.plot(top_vendors['VendorName'],
         top_vendors['Cumulative_Contribution%'],
         color='red', marker='o', linestyle='dashed', label='Cumulative␣
 ↪Contribution %')

# Fix x-axis labels
ax1.set_xticks(range(len(top_vendors['VendorName'])))
ax1.set_xticklabels(top_vendors['VendorName'], rotation=90)

# Axis labels and title
ax1.set_ylabel('Purchase Contribution %', color='blue')
ax2.set_ylabel('Cumulative Contribution %', color='red')
ax1.set_xlabel('Vendors')
ax1.set_title('Pareto Chart: Vendor Contribution to Total Purchases')

# Rescale y-axes for better visibility
```

```
ax1.set_ylim(0, top_vendors['Purchase_Contribution%'].max() + 1)
ax2.set_ylim(0, 100)   # cumulative contribution should go to 100%

# Horizontal line at 100%
ax2.axhline(y=100, color='gray', linestyle='dashed', alpha=0.7)

# Legend
lines, labels = ax2.get_legend_handles_labels()
ax2.legend(lines, labels, loc='upper right')

plt.tight_layout()
plt.show()
```



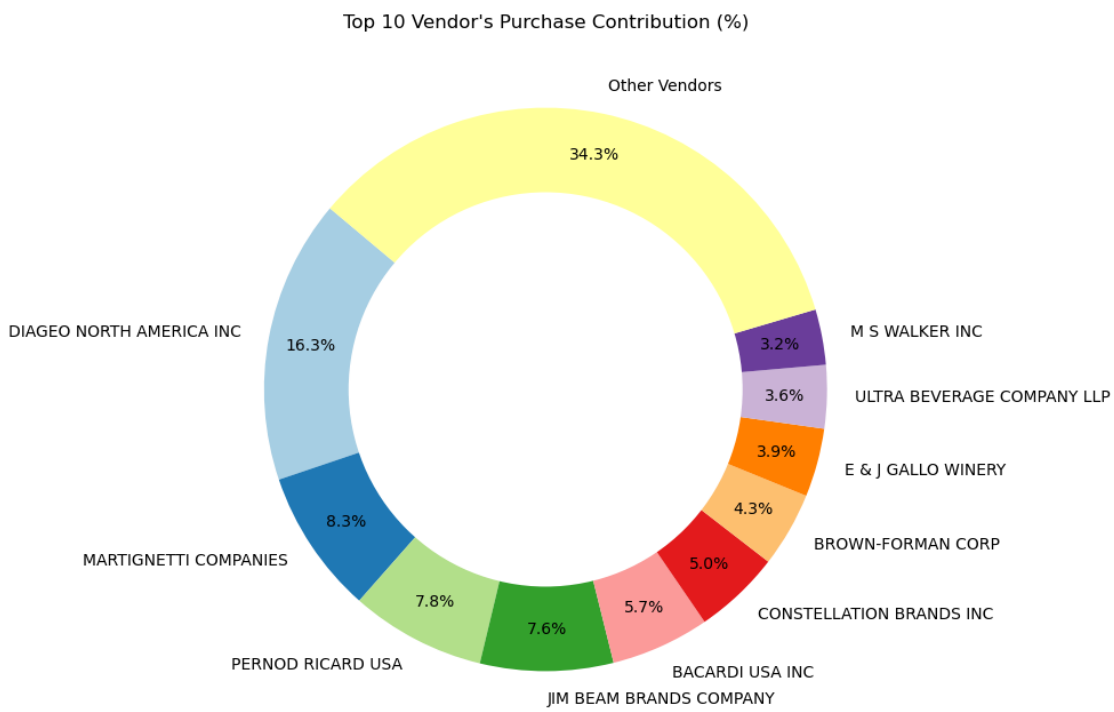
2.1.1 How much of total procurment is dependant on the top 10 vendors ?

[125]:
```
print(f"Total Contribution of the top 10 vendors is {round(␣
↪top_vendors['Purchase_Contribution%'].sum(),2)}%")
```

```
Total Contribution of the top 10 vendors is 65.69%
```

[126]:
```
vendors = list(top_vendors['VendorName'].values)
purchase_contributions = list(top_vendors['Purchase_Contribution%'].values)
total_contribution = sum(purchase_contributions)
remaining_contribution = 100 - total_contribution

# Append "Other Vendors" category
```

```
vendors.append("Other Vendors")
purchase_contributions.append(remaining_contribution)

# Donut Chart
fig, ax = plt.subplots(figsize=(8, 8))
wedges, texts, autotexts = ax.pie(purchase_contributions, labels=vendors,␣
↪autopct='%1.1f%%',
                                  startangle=140, pctdistance=0.85, colors=plt.
↪cm.Paired.colors)

# Draw a white circle in the center to create a "donut" effect
centre_circle = plt.Circle((0, 0), 0.70, fc='white')
fig.gca().add_artist(centre_circle)

plt.title("Top 10 Vendor's Purchase Contribution (%)")
plt.show()
```



2.2 Key Insights

2.2.1 1. Highly Concentrated Vendor Influence

- **DIAGEO NORTH AMERICA INC** alone accounts for **16.3%** of total purchases — the largest single share.

- The top 3 vendors (DIAGEO, MARTIGNETTI, PERNOD RICARD) together contribute over **32%**, indicating a strong reliance on a few dominant suppliers.

2.2.2 2. Long Tail of Smaller Vendors

- **"Other Vendors"** make up **34.3%**, suggesting a wide base of smaller suppliers.
- This could reflect niche sourcing, regional diversity, or fragmented purchasing for specialized products.

2.2.3 3. Strategic Vendor Relationships

- Vendors like **JIM BEAM**, **BACARDI**, and **CONSTELLATION BRANDS** each contribute between **5–8%**, showing they're significant but not dominant.
- These mid-tier suppliers may offer leverage for negotiation or diversification.

2.3 Business Implications

- **Risk Management**: Heavy reliance on top vendors could pose supply chain risks. Diversifying or securing long-term contracts may be wise.
- **Negotiation Power**: High-volume vendors may offer better pricing or terms — but also hold more influence.
- **Opportunity Areas**: Smaller vendors (in the "Other" category) might be explored for innovation, local sourcing, or cost savings.

2.3.1 Does purchasing in bulk reduce the unit price and what is the optimal purchase volume for cost saving ?

[127]:
```
df['UnitPurchasePrice']=df['TotalPurchaseDollars']/df['TotalPurchaseQuantity']
```

[128]:
```
df
```

[128]:
```
      VendorNumber                VendorName  Brand  \
0             1128         BROWN-FORMAN CORP   1233
1             4425     MARTIGNETTI COMPANIES   3405
2            17035         PERNOD RICARD USA   8068
3             3960  DIAGEO NORTH AMERICA INC   4261
4             3960  DIAGEO NORTH AMERICA INC   3545
…             …                        …     …
8559          9815            WINE GROUP INC   8527
8560          8004            SAZERAC CO INC   5683
8561          3924  HEAVEN HILL DISTILLERIES   9123
8562          3960  DIAGEO NORTH AMERICA INC   6127
8563          7245      PROXIMO SPIRITS INC.   3065

                      Description  PurchasePrice  ActualPrice  Volume  \
0          Jack Daniels No 7 Black          26.27        36.99  1750.0
1            Tito's Handmade Vodka          23.19        28.99  1750.0
```

2	Absolut 80 Proof	18.24	24.99	1750.0
3	Capt Morgan Spiced Rum	16.17	22.99	1750.0
4	Ketel One Vodka	21.89	29.99	1750.0
…	…	…	…	…
8559	Concannon Glen Ellen Wh Zin	1.32	4.99	750.0
8560	Dr McGillicuddy's Apple Pie	0.39	0.49	50.0
8561	Deep Eddy Vodka	0.74	0.99	50.0
8562	The Club Strawbry Margarita	1.47	1.99	200.0
8563	Three Olives Grape Vodka	0.71	0.99	50.0

	TotalPurchaseQuantity	TotalPurchaseDollars	TotalSalesQuantity	\
0	145080	3811251.60	142049.0	
1	164038	3804041.22	160247.0	
2	187407	3418303.68	187140.0	
3	201682	3261197.94	200412.0	
4	138109	3023206.01	135838.0	
…	…	…	…	
8559	2	2.64	5.0	
8560	6	2.34	134.0	
8561	2	1.48	2.0	
8562	1	1.47	72.0	
8563	1	0.71	86.0	

	TotalSalesDollars	TotalSalesPrice	ExciseTax	FreightCost	GrossProfit	\
0	5101919.51	672819.31	260999.20	68601.68	1290667.91	
1	4819073.49	561512.37	294438.66	144929.24	1015032.27	
2	4538120.60	461140.15	343854.07	123780.22	1119816.92	
3	4475972.88	420050.01	368242.80	257032.07	1214774.94	
4	4223107.62	545778.28	249587.83	257032.07	1199901.61	
…	…	…	…	…	…	
8559	15.95	10.96	0.55	27100.41	13.31	
8560	65.66	1.47	7.04	50293.62	63.32	
8561	1.98	0.99	0.10	14069.87	0.50	
8562	143.28	77.61	15.12	257032.07	141.81	
8563	85.14	33.66	4.46	38994.78	84.43	

	ProfitMargin	SalesTurnover	SalestoPurchaseRatio	UnitPurchasePrice
0	25.297693	0.979108	1.338647	26.27
1	21.062810	0.976890	1.266830	23.19
2	24.675786	0.998575	1.327594	18.24
3	27.139908	0.993703	1.372493	16.17
4	28.412764	0.983556	1.396897	21.89
…	…	…	…	…
8559	83.448276	2.500000	6.041667	1.32
8560	96.436186	22.333333	28.059829	0.39
8561	25.252525	1.000000	1.337838	0.74
8562	98.974037	72.000000	97.469388	1.47

```
8563     99.166079      86.000000            119.915493                0.71

[8564 rows x 19 columns]
```

[129]:
```
df['OrderSize']= pd.
↪qcut(df['TotalPurchaseQuantity'],q=3,labels=['Small','Medium','Large'])
```

[130]:
```
df['OrderSize']
```

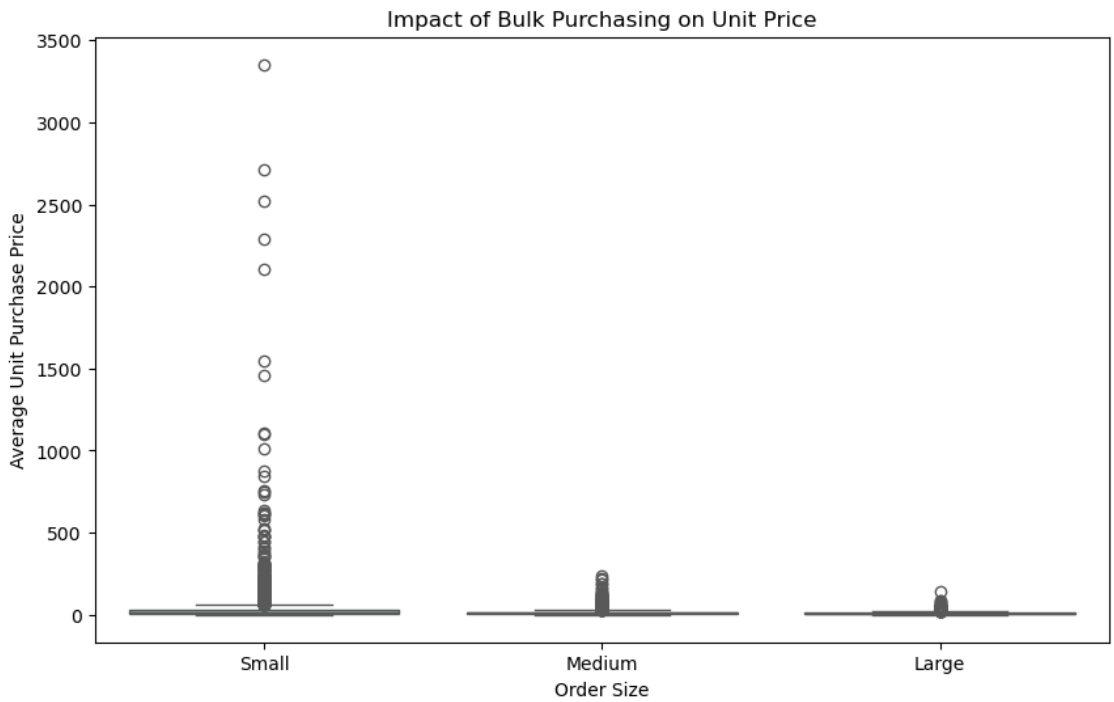
[130]:
```
0       Large
1       Large
2       Large
3       Large
4       Large
        …
8559    Small
8560    Small
8561    Small
8562    Small
8563    Small
Name: OrderSize, Length: 8564, dtype: category
Categories (3, object): ['Small' < 'Medium' < 'Large']
```

[131]:
```
df.groupby('OrderSize')[['UnitPurchasePrice']].mean()
```

[131]:
```
           UnitPurchasePrice
OrderSize
Small              39.068186
Medium             15.486414
Large              10.777625
```

[132]:
```
# Set the figure size to 10 inches wide and 6 inches tall
plt.figure(figsize=(10, 6))

# Create a boxplot showing distribution of UnitPurchasePrice across OrderSize␣
↪categories
# 'data=df' specifies the DataFrame
# 'x="OrderSize"' sets the categorical variable on the x-axis
# 'y="UnitPurchasePrice"' sets the numeric variable on the y-axis
# 'palette="Set2"' applies a pastel color scheme


sns.boxplot(data=df, x="OrderSize", y="UnitPurchasePrice", palette="Set2")

# Add a title to the plot
plt.title("Impact of Bulk Purchasing on Unit Price")
```

```
# Label the x-axis
plt.xlabel("Order Size")

# Label the y-axis
plt.ylabel("Average Unit Purchase Price")

# Display the plot
plt.show()
```



- Vendors buying in bulk (Large Order Size) get the lowest unit price ($10.78 per unit), meaning higher margins if they can manage inventory efficiently.

- The price difference between Small and Large orders is substantial (~72% reduction in unit cost).

- This suggests that bulk pricing strategies successfully encourage vendors to purchase in larger volumes, leading to higher overall sales despite lower per-unit revenue.

2.4 Key Observations

2.4.1 1. Smaller Orders Have Higher Price Variability

- The **"Small" Order Size** category shows a **wide spread** in unit prices.
- There are **many outliers**, some reaching up to **3300**, suggesting inconsistent pricing for small purchases.

- This could indicate less negotiation power or more frequent one-off purchases.

2.4.2 2. Bulk Purchasing Reduces Unit Price

- Both **"Medium"** and **"Large"** order sizes have **lower median unit prices** compared to "Small".
- The boxes (IQR) are **narrower**, indicating more consistent pricing.
- This supports the idea that **buying in bulk leads to cost efficiency**.

2.4.3 3. Outlier Behavior

- Outliers are **much more frequent** in the "Small" category.
- "Medium" and "Large" have **fewer or no extreme outliers**, suggesting more standardized pricing practices.

2.5 Business Implications

- **Encourage bulk purchasing** to stabilize and reduce unit costs.
- **Review pricing strategy** for small orders — consider minimum order thresholds or bundled pricing.
- **Segment customers** by order size to tailor pricing and incentives.

Which Vendors have low inventory turnover, indicating excess stock and slow moving product ?

[134]:
```
# Column name should be StockTurnover NOT SalesTurnover
df.rename(columns={'SalesTurnover': 'StockTurnover'}, inplace=True)
```

[137]:
```
df[df['StockTurnover'] < 1].groupby('VendorName')[['StockTurnover']].mean().
↪sort_values('StockTurnover',ascending=True).head(10)
```

[137]:

VendorName	StockTurnover
ALISA CARR BEVERAGES	0.615385
HIGHLAND WINE MERCHANTS LLC	0.708333
PARK STREET IMPORTS LLC	0.751306
Circa Wines	0.755676
Dunn Wine Brokers	0.766022
CENTEUR IMPORTS LLC	0.773953
SMOKY QUARTZ DISTILLERY LLC	0.783835
TAMWORTH DISTILLING	0.797078
THE IMPORTED GRAPE LLC	0.807569
WALPOLE MTN VIEW WINERY	0.820548

2.5.1 Key Observations

- **Low Stock Turnover Values**: All vendors listed have stock turnover ratios below 1.0, which typically indicates **slow inventory movement**. This means products are not being sold or replaced quickly, potentially leading to higher holding costs or outdated stock.

- **Ranking by Efficiency**: The vendors are sorted in ascending order of turnover, so:
  - **ALISA CARR BEVERAGES** has the **lowest turnover** (0.615), suggesting the slowest inventory movement.
  - **WALPOLE MTN VIEW WINERY** has the **highest turnover** in this list (0.820), indicating relatively better inventory flow.
- **Operational Implications**:
  - Vendors with **higher turnover** (closer to 1.0) may be managing inventory more efficiently, possibly due to better demand forecasting, faster sales cycles, or leaner stock levels.
  - Vendors with **lower turnover** might need to investigate issues like overstocking, poor sales performance, or supply chain delays.

2.6 How much capital is tied up in unslod inventory ?

[144]:
```
# Total Unsold Inventory Value across all Vendors
df['UnsoldInventoryValue'] = (df['TotalPurchaseQuantity'] -␣
↪df['TotalSalesQuantity'])*df['PurchasePrice']

print('Total Unsold Capital',format_dollars(df['UnsoldInventoryValue'].sum()))
```

```
Total Unsold Capital 2.71M
```

[147]:
```
# Unsold Inventory Capital per Vendor
inventrory_value_per_vendor = df.groupby("VendorName")['UnsoldInventoryValue'].
↪sum().reset_index()
inventrory_value_per_vendor
```

[147]:
```
                      VendorName  UnsoldInventoryValue
0        ADAMBA IMPORTS INTL INC                 -91.52
1           ALISA CARR BEVERAGES              -43151.48
2             ALTAMAR BRANDS LLC                 464.23
3      AMERICAN SPIRITS EXCHANGE                 116.76
4      AMERICAN VINTAGE BEVERAGE                3672.17
..                           …                    …
114               WEIN BAUER INC                1732.03
115  WESTERN SPIRITS BEVERAGE CO                -949.61
116     WILLIAM GRANT & SONS INC              401960.83
117               WINE GROUP INC               86862.57
118            ZORVINO VINEYARDS                2864.26

[119 rows x 2 columns]
```

[157]:
```
# Aggregate Capital Locked per Vendor
inventory_value_per_vendor = df.groupby("VendorName")["UnsoldInventoryValue"].
↪sum().reset_index()
```

```
# Sort Vendors with the Highest Locked Capital
inventory_value_per_vendor = inventory_value_per_vendor.
 ↪sort_values(by="UnsoldInventoryValue", ascending=False)
inventory_value_per_vendor["UnsoldInventoryValue"] =␣
 ↪inventory_value_per_vendor["UnsoldInventoryValue"].apply(format_dollars)
inventory_value_per_vendor.head(10)
```

[157]:

```
                    VendorName UnsoldInventoryValue
25   DIAGEO NORTH AMERICA INC              722.21K
46    JIM BEAM BRANDS COMPANY              554.67K
68          PERNOD RICARD USA              470.63K
116  WILLIAM GRANT & SONS INC              401.96K
30         E & J GALLO WINERY              228.28K
79             SAZERAC CO INC              198.44K
11          BROWN-FORMAN CORP              177.73K
20   CONSTELLATION BRANDS INC              133.62K
61      MOET HENNESSY USA INC              126.48K
77     REMY COINTREAU USA INC              118.60K
```

2.7 What is the 95% confidence interval for the profit margins of top-performing and bottom-performing vendors

[158]:

```
top_threshold = df['TotalSalesDollars'].quantile(0.75)
low_threshold = df['TotalSalesDollars'].quantile(0.25)
```

[159]:

```
top_vendors = df[df["TotalSalesDollars"] >= top_threshold]["ProfitMargin"].
 ↪dropna()
low_vendors = df[df["TotalSalesDollars"] <= low_threshold]["ProfitMargin"].
 ↪dropna()
```

[163]:

```
top_vendors

low_vendors
```

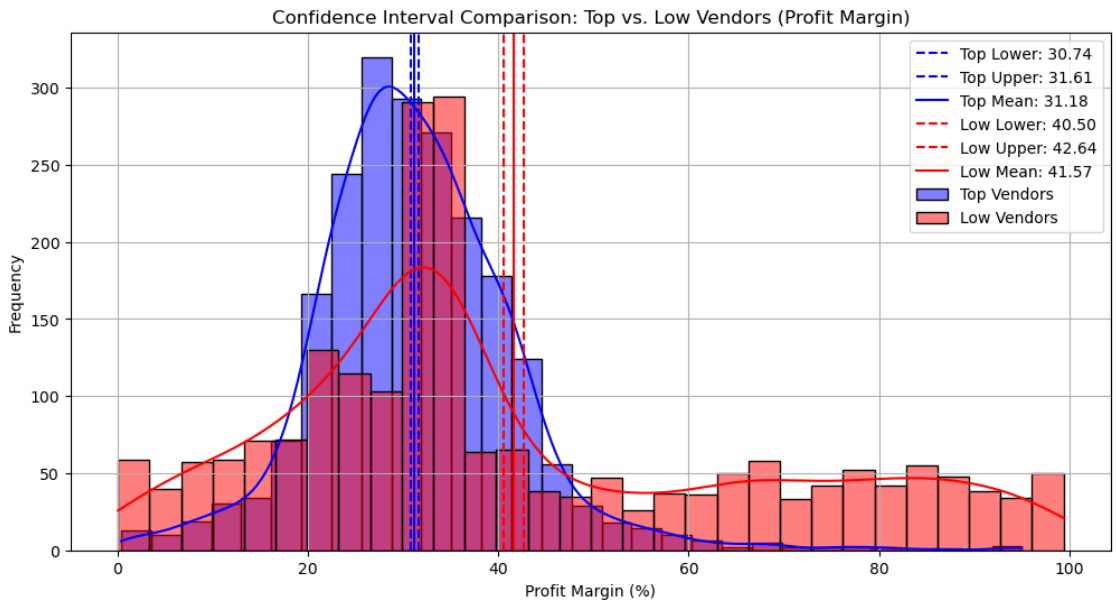
[163]:

```
5631     4.111764
5652     6.145626
5701    12.007271
5704     1.677308
5724     7.239599
          …
8559    83.448276
8560    96.436186
8561    25.252525
8562    98.974037
8563    99.166079
Name: ProfitMargin, Length: 2141, dtype: float64
```

[164]:
```
def confidence_interval(data, confidence=0.95):
    mean_val = np.mean(data)
    std_err = np.std(data, ddof=1) / np.sqrt(len(data))  # Standard error
    t_critical = stats.t.ppf((1 + confidence) / 2, df=len(data) - 1)
    margin_of_error = t_critical * std_err
    return mean_val, mean_val - margin_of_error, mean_val + margin_of_error
```

[165]:
```
top_mean, top_lower, top_upper = confidence_interval(top_vendors)
low_mean, low_lower, low_upper = confidence_interval(low_vendors)

print(f"Top Vendors 95% CI: {top_lower:.2f}, {top_upper:.2f}, Mean: {top_mean:.
 ↪2f}")
print(f"Low Vendors 95% CI: {low_lower:.2f}, {low_upper:.2f}, Mean: {low_mean:.
 ↪2f}")

plt.figure(figsize=(12, 6))

# Top Vendors Plot
sns.histplot(top_vendors, kde=True, color="blue", bins=30, alpha=0.5,␣
 ↪label="Top Vendors")
plt.axvline(top_lower, color="blue", linestyle="--", label=f"Top Lower:␣
 ↪{top_lower:.2f}")
plt.axvline(top_upper, color="blue", linestyle="--", label=f"Top Upper:␣
 ↪{top_upper:.2f}")
plt.axvline(top_mean, color="blue", linestyle="-", label=f"Top Mean: {top_mean:.
 ↪2f}")

# Low Vendors Plot
sns.histplot(low_vendors, kde=True, color="red", bins=30, alpha=0.5, label="Low␣
 ↪Vendors")
plt.axvline(low_lower, color="red", linestyle="--", label=f"Low Lower:␣
 ↪{low_lower:.2f}")
plt.axvline(low_upper, color="red", linestyle="--", label=f"Low Upper:␣
 ↪{low_upper:.2f}")
plt.axvline(low_mean, color="red", linestyle="-", label=f"Low Mean: {low_mean:.
 ↪2f}")

# Finalize Plot
plt.title("Confidence Interval Comparison: Top vs. Low Vendors (Profit Margin)")
plt.xlabel("Profit Margin (%)")
plt.ylabel("Frequency")
plt.legend()
plt.grid(True)
plt.show()
```

```
Top Vendors 95% CI: 30.74, 31.61, Mean: 31.18
Low Vendors 95% CI: 40.50, 42.64, Mean: 41.57
```



2.7.1 Key Observations from the Graph

1. Profit Margin Distributions

- **Top Vendors (Blue)**:
  - Mean Profit Margin: **31.18%**
  - 95% Confidence Interval: **30.74% to 31.61%**
- **Low Vendors (Red)**:
  - Mean Profit Margin: **41.57%**
  - 95% Confidence Interval: **40.50% to 42.64%**

This shows that **low-performing vendors have significantly higher profit margins** than top-performing ones.

2.7.2 What This Suggests

Inverse Relationship Between Sales and Margin

- Vendors with **lower total sales** tend to have **higher profit margins**.
- Vendors with **higher total sales** operate on **thinner margins** — possibly due to volume-based pricing, competitive pressure, or economies of scale.

Statistical Significance

- The confidence intervals **do not overlap**, which strongly suggests the difference in means is **statistically significant**.
- This means the observed difference is unlikely due to random chance.

Distribution Shape

- Both groups show smooth KDE curves, indicating reasonably normal distributions.
- The top vendors' distribution is tighter (narrower CI), suggesting **less variability** in their profit margins.
- Low vendors show a slightly wider spread, indicating **more variability**.

2.7.3 Strategic Implications

- **High-margin, low-volume vendors** might be niche or premium sellers.
- **Low-margin, high-volume vendors** could be mainstream or discount-driven.
- This insight could guide pricing strategy, vendor segmentation, or performance benchmarking.

2.8 Hypothesis Test: Profit Margin Comparison

Question:
Is there a significant difference in profit margins between top-performing and low-performing vendors?

2.8.1 Hypotheses

- **H (Null Hypothesis):**
  There is no significant difference in the mean profit margins of top-performing and low-performing vendors.
- **H (Alternative Hypothesis):**
  The mean profit margins of top-performing and low-performing vendors are significantly different.}

[171]:
```
top_threshold = df["TotalSalesDollars"].quantile(0.75)
low_threshold = df["TotalSalesDollars"].quantile(0.25)

top_vendors = df[df["TotalSalesDollars"] >= top_threshold]["ProfitMargin"].
 ↪dropna()
low_vendors = df[df["TotalSalesDollars"] <= low_threshold]["ProfitMargin"].
 ↪dropna()

# Perform Two-Sample T-Test
t_stat, p_value = ttest_ind(top_vendors, low_vendors, equal_var=False)

# Print results
print(f"T-Statistic: {t_stat:.4f}, P-Value: {p_value:.4f}")
if p_value < 0.05:
    print("Reject Ho: There is a significant difference in profit margins␣
 ↪between top and low-performing vendors.")
else:
    print("Fail to Reject Ho: No significant difference in profit margins.")
```

```
T-Statistic: -17.6695, P-Value: 0.0000
Reject Ho: There is a significant difference in profit margins between top and
```

low-performing vendors.

[172]:
```
df.to_csv("vendors_data.csv", index=False)
```

[ ]:
```

```